(134) (2016-SE-01)

process P	process Q	
p.1	q.1	⇝ p.1 преди q.2
p.2	q.2	и
p.3	q.3	q.2 преди p.3

Решение: sem.1 := 0 , sem.2 := 0

process P	process Q
p.1	q.1
signal(sem.1)	wait(sem.1)
p.2	q.2
wait(sem.2)	signal(sem.2)
p.3	q.3

(135) (2017-SE-01)

process P	process Q	process R	
p.1	q.1	r.1	⇝ p.1 преди q.2 и
p.2	q.2	r.2	r.2
p.3	q.3	r.3	

Решение: sem.1 := 0

process P	process Q	process R
p.1	q.1	r.1
signal(sem.1)	wait(sem.1)	wait(sem.1)
signal(sem.1)	q.2	p.2
p.2	q.3	p.3
p.3		

(136) (2017-SE-04)

process P	process Q	
p_1	q_1	⇝ p_1 преди q_2
p_2	q_2	q_1 преди p_2

Решение: sem_1:=0 , sem_2:=0

process P	process Q
p_1	q_1
signal(sem_1)	wait(sem_1)
wait(sem_2)	signal(sem_2)
p_2	q_2

(137) (2017-SE-05)

process P	process Q	
p_1	q_1	⇝ p_1, q_1, p_2, q_2, p_3, q_3
p_2	q_2	
p_3	q_3	

Решение: sem_1:=0 , sem_2:=0

process P	process Q
wait(sem_2)	wait(sem_1)
p_1	q_1
signal(sem_1)	signal(sem_2)
wait(sem_2)	wait(sem_1)
p_2	q_2
signal(sem_1)	signal(sem_2)
wait(sem_2)	wait(sem_1)
p_3	q_3
signal(sem_1)	

(1140) (2018-SE-01)

process P	process Q	process R
p.1	q.1	r.1
p.2	q.2	r.2
p.3	q.3	r.3

p.2 или q.2 преди r.2

Решение: ~~sem.1:=0, sem.2:=0~~

~~process P~~	~~process Q~~	~~process R~~
~~p.1~~	~~q.1~~	~~r.1~~
~~sig~~	~~wait(sem.2)~~	~~wait(sem.1)~~
~~p.2~~	~~q.2~~	~~r.2~~
~~signal(sem.1)~~	~~q.3~~	~~signal(sem.2)~~
~~p.3~~		~~r.3~~

mutex:=1 , Barrier:=0

p.1	q.1	r.1
wait(mutex)	wait(mutex)	wait(Barrier)
p.2	q.2	r.2
signal(Barrier)	signal(mutex)	signal(mutex)
p.3	q.3	r.3

(145) (2020-SE-05)

Решение:

~~p-1 signal(sem-1) signal(sem-2)~~
~~q-1 wait(sem-1) q-2~~
~~r-1 wait(sem-2) r-2~~

sem-1 := 0 sem-2 := 1

p-1	q-1	r-1
signal(sem-1)	wait(sem-1)	wait(sem-1)
signal(sem-1	wait(sem-2)	wait(sem-2)
p-2	q-2	r-2
p-3	q-3	r-3
	signal(sem-2)	signal(sem-2)

(146) (2025-SE-01)

Решение: mutex := 1, empty := N, full := 0

```
push(s,x):
  wait(s.empty)
  wait(s.mutex)
  s.buf[s.idx] := x
  s.idx += 1
  signal(s.mutex)
  signal(s.full)
```

```
pop(s):
  wait(s.full)
  wait(s.mutex)
  result := s.buf[s.idx - 1]
  s.idx -= 1
  signal(s.mutex)
  signal(s.empty)
  return result
```